

Diagnostic report: Mg I

Computed levels: 192
Matched to NIST: 39
Computed E1 lines: 9999
Level-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 90.7 cm⁻¹

Energy-level fit vs Bob Kurucz ($E_{\text{obs}} \leq 60000$ cm⁻¹, offset-removed):

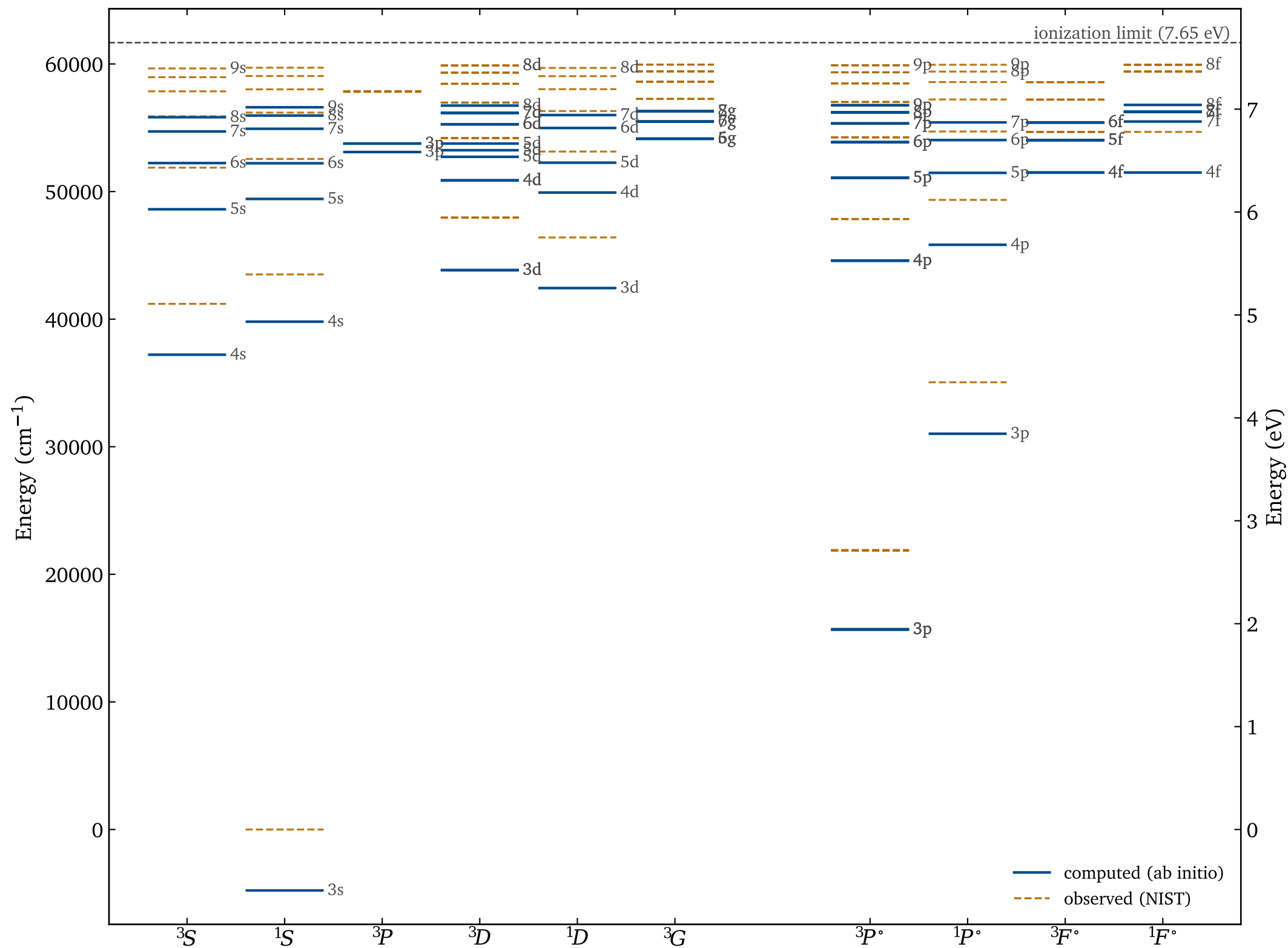
	N	median E-0	RMS	90th	max
this fit	102	3.0	91	77	606
Bob (RCE)	118	0.0	18	22	78

bulk (median) is near-Bob; the ~3x RMS gap is the singlet
3s.nd 1D / near-IE tail (unobserved-perturber wall), not
data starvation -- the basis matches Bob's b1200{e,o} deck.

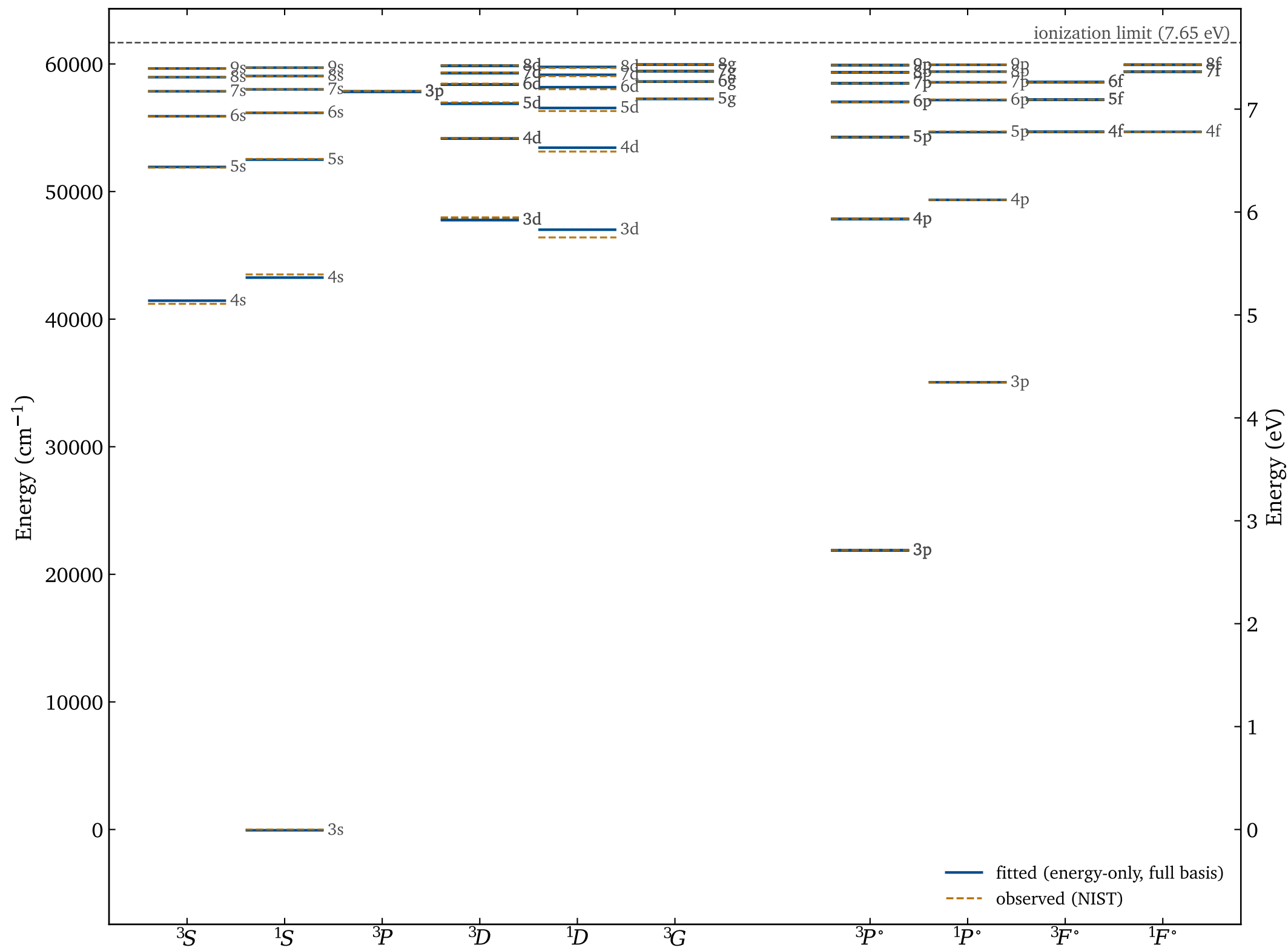
Strongest computed lines:

lambda(A)	log gf	upper term
123623.27	+1.406	(2S) 3H
190473.12	+1.360	(2S) 3H
123623.27	+1.338	(2S) 1H
123622.58	+1.337	(2S) 3H
277849.65	+1.330	(2S) 3H
190473.12	+1.292	(2S) 1H
190472.30	+1.291	(2S) 3H
277849.65	+1.262	(2S) 1H

Mg I: term diagram - ab initio vs observed



Mg I: term diagram - fitted vs observed



Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s2	¹ S	0	0.0	-4776.0	-4776.0	-57.8	-57.8
3s.3p	³ P°	0	21850.4	15663.0	-6187.4	21860.2	+9.8
3s.3p	³ P°	1	21870.5	15676.0	-6194.5	21873.2	+2.8
3s.3p	³ P°	2	21911.2	15702.0	-6209.2	21900.2	-10.9
3s.3p	¹ P°	1	35051.3	31018.0	-4033.3	35051.2	-0.0
3s.4s	³ S	1	41197.4	37213.0	-3984.4	41445.2	+247.8
3s.4s	¹ S	0	43503.3	39801.0	-3702.3	43254.2	-249.1
3s.3d	¹ D	2	46403.1	42435.0	-3968.1	47009.2	+606.2
3s.4p	³ P°	0	47841.1	44585.0	-3256.1	47841.2	+0.1
3s.4p	³ P°	1	47844.4	44588.0	-3256.4	47844.2	-0.2
3s.4p	³ P°	2	47851.2	44593.0	-3258.2	47849.2	-1.9
3s.3d	³ D	1	47957.1	43839.0	-4118.1	47773.2	-183.8
3s.3d	³ D	2	47957.0	43839.0	-4118.0	47774.2	-182.8
3s.3d	³ D	3	47957.0	43839.0	-4118.0	47774.2	-182.8
3s.4p	¹ P°	1	49346.7	45828.0	-3518.7	49348.2	+1.5
3s.5s	³ S	1	51872.5	48610.0	-3262.5	51926.2	+53.7
3s.5s	¹ S	0	52556.2	49421.0	-3135.2	52499.2	-57.0
3s.4d	¹ D	2	53134.6	49922.0	-3212.6	53439.2	+304.6
3s.4d	³ D	3	54192.3	50873.0	-3319.3	54148.2	-44.0
3s.4d	³ D	2	54192.3	50873.0	-3319.3	54148.2	-44.0
3s.4d	³ D	1	54192.3	50873.0	-3319.3	54148.2	-44.1
3s.5p	³ P°	0	54248.8	51075.0	-3173.8	54265.2	+16.4
3s.5p	³ P°	1	54250.1	51076.0	-3174.1	54266.2	+16.2
3s.5p	³ P°	2	54252.7	51078.0	-3174.7	54269.2	+16.5
3s.4f	³ F°	2	54676.7	51486.0	-3190.7	54676.2	-0.4
3s.4f	³ F°	3	54676.7	51486.0	-3190.7	54676.2	-0.5
3s.4f	³ F°	4	54676.8	51486.0	-3190.8	54676.2	-0.5
3s.4f	¹ F°	3	54676.4	51492.0	-3184.4	54680.2	+3.8
3s.5p	¹ P°	1	54706.5	51464.0	-3242.5	54658.2	-48.3
3s.6s	³ S	1	55891.8	52232.0	-3659.8	55902.2	+10.4
3s.6s	¹ S	0	56186.9	52216.0	-3970.9	56169.2	-17.6
3s.5d	¹ D	2	56308.4	52262.0	-4046.4	56550.2	+241.9
3s.5d	³ D	3	56968.2	53761.0	-3207.2	56891.2	-77.0

Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s.5d	3D	1	56968.3	52721.0	-4247.3	56891.2	-77.0
3s.6p	$^3P^\circ$	0	57017.1	53866.0	-3151.1	57031.2	+14.2
3s.6p	$^3P^\circ$	1	57017.7	53867.0	-3150.7	57032.2	+14.5
3s.6p	$^3P^\circ$	2	57019.0	53868.0	-3151.0	57033.2	+14.2
3s.6p	$^1P^\circ$	1	57215.0	54034.0	-3181.0	57172.2	-42.7
3s.5f	$^3F^\circ$	2	57204.2	54029.0	-3175.2	57204.2	+0.0
3s.5f	$^3F^\circ$	3	57204.3	54029.0	-3175.3	57204.2	-0.0
3s.5f	$^3F^\circ$	4	57204.3	54029.0	-3175.3	57204.2	-0.1
3s.5f	$^3F^\circ$	3	57204.2	54034.0	-3170.2	57206.2	+2.1
3s.5g	3G	4	57262.8	54140.0	-3122.8	57263.2	+0.5
3s.5g	3G	3	57262.8	54140.0	-3122.8	57263.2	+0.5
3p2	3P	0	57812.8	53098.0	-4714.8	57813.2	+0.5
3p2	3P	1	57833.4	53761.0	-4072.4	57828.2	-5.2
3p2	3P	2	57873.9	53761.0	-4112.9	57859.2	-14.7
3s.7s	3S	1	57855.2	54709.0	-3146.2	57858.2	+3.0
3s.7s	1S	0	58009.4	54919.0	-3090.4	58007.2	-2.2
3s.6d	1D	2	58023.2	54982.0	-3041.2	58179.2	+156.0
3s.6d	3D	3	58442.8	55274.0	-3168.8	58390.2	-52.6
3s.6d	3D	2	58442.9	55274.0	-3168.9	58390.2	-52.6
3s.6d	3D	1	58442.9	55274.0	-3168.9	58390.2	-52.6
3s.7p	$^3P^\circ$	0	58476.7	55334.0	-3142.7	58483.2	+6.6
3s.7p	$^3P^\circ$	1	58477.0	55334.0	-3143.0	58483.2	+6.2
3s.7p	$^3P^\circ$	2	58477.8	55335.0	-3142.8	58484.2	+6.5
3s.7p	$^1P^\circ$	1	58580.2	55420.0	-3160.2	58566.2	-14.0
3s.6f	$^3F^\circ$	2	58575.5	55416.0	-3159.5	58575.2	-0.3
3s.6f	$^3F^\circ$	3	58575.5	55416.0	-3159.5	58575.2	-0.3
3s.6f	$^3F^\circ$	4	58575.5	55416.0	-3159.5	58575.2	-0.3
3s.6f	$^3F^\circ$	3	58575.5	55419.0	-3156.5	58577.2	+1.8
3s.6g	3G	4	58610.8	54140.0	-4470.8	58611.2	+0.5
3s.6g	3G	3	58610.8	55482.0	-3128.8	58611.2	+0.5
3s.6g	3G	5	58610.8	54140.0	-4470.8	58611.2	+0.5
3s.8s	3S	1	58962.7	55824.0	-3138.7	58962.2	-0.5
3s.8s	1S	0	59053.5	55956.0	-3097.5	59055.2	+1.7

Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s.7d	3D	3	59318.8	56166.0	-3152.8	59282.2	-36.5
3s.7d	3D	2	59318.8	56166.0	-3152.8	59282.2	-36.5
3s.7d	3D	1	59318.8	56166.0	-3152.8	59282.2	-36.5
3s.8p	$^3P^\circ$	0	59342.5	56203.0	-3139.5	59345.2	+2.7
3s.8p	$^3P^\circ$	1	59342.5	56203.0	-3139.5	59345.2	+2.7
3s.8p	$^3P^\circ$	2	59342.5	56204.0	-3138.5	59345.2	+2.7
3s.8p	$^1P^\circ$	1	59403.2	--	--	59396.2	-6.9
3s.7f	$^1F^\circ$	2	59400.8	56251.0	-3149.8	59400.2	-0.5
3s.7f	$^1F^\circ$	3	59400.8	56251.0	-3149.8	59400.2	-0.5
3s.7f	$^1F^\circ$	4	59400.8	55487.0	-3913.8	59401.2	+0.5
3s.7f	$^1F^\circ$	3	59400.8	56253.0	-3147.8	59402.2	+1.5
3s.7g	3G	3	59423.5	56292.0	-3131.5	59423.2	-0.3
3s.7g	3G	4	59423.5	55482.0	-3941.5	59423.2	-0.3
3s.7g	3G	5	59423.5	55482.0	-3941.5	59423.2	-0.3
3s.7g	3G	4	59423.5	55482.0	-3941.5	59424.2	+0.7
3s.9s	3S	1	59649.2	--	--	59646.2	-2.9
3s.9s	1S	0	59707.1	56604.0	-3103.1	59711.2	+4.1
3s.8d	1D	2	59690.0	--	--	59765.2	+75.3
3s.8d	3D	3	59881.2	56737.0	-3144.2	59857.2	-23.9
3s.8d	3D	2	59881.2	--	--	59857.2	-23.9
3s.8d	3D	1	59881.2	--	--	59857.2	-24.0
3s.9p	$^3P^\circ$	0	59897.9	56761.0	-3136.9	59899.2	+1.4
3s.9p	$^3P^\circ$	1	59897.9	--	--	59899.2	+1.4
3s.9p	$^3P^\circ$	2	59897.9	56761.0	-3136.9	59899.2	+1.4
3s.9p	$^1P^\circ$	1	59936.6	--	--	59934.2	-2.4
3s.8f	$^1F^\circ$	2	59935.4	--	--	59936.2	+0.9
3s.8f	$^1F^\circ$	3	59935.4	56791.0	-3144.4	59936.2	+0.9
3s.8f	$^1F^\circ$	4	59935.4	56251.0	-3684.4	59936.2	+0.9
3s.8g	3G	4	59950.9	56292.0	-3658.9	59951.2	+0.4
3s.8g	3G	5	59950.9	56292.0	-3658.9	59951.2	+0.4

